CS19541 – COMPUTER NETWORKS LAB

PRACTICAL 11 – STATIC ROUTING AND RIP

AIM

To set up and check static routing and RIP routing between networks with Cisco Packet Tracer.

REQUIREMENTS

Cisco Packet Tracer, routers, PCs, and suitable Ethernet and serial cables.

PROCEDURE

1. Add three routers and the necessary PCs to the Packet Tracer workspace.

2. Connect the devices with the appropriate cables.

3. Configure IP addresses on router interfaces and PCs, and set the default gateways.

4. Activate the router interfaces with the no shutdown command. If needed, set the clock rate on DCE serial interfaces.

5. Create static routes with the ip route command and configure primary and backup paths where required.

6. Check the static routing entries using show ip route static.

7. Start RIP with router rip and announce directly connected networks using the network commands.

8. Check communication using ping and tracert.

SAMPLE STATIC ROUTE

```
ip route [destination-network] [subnet-mask] [next-hop-IP]
```

SAMPLE RIP CONFIGURATION

```
router rip
version 2
network [network-address]
no auto-summary
```

INPUT

The required inputs are network addresses, subnet masks, next-hop IP addresses, and router interface settings.

OUTPUT

The routing tables show the configured static routes or routes learned through RIP. The ping and tracert commands are used to verify connectivity and the route between PCs.

OBSERVATION

1. The three routers and required PCs were connected in Cisco Packet Tracer, and IP addresses were configured for all interfaces and PCs.

2. Router interfaces were enabled with no shutdown. Suitable clock-rate and bandwidth values were set on DCE serial interfaces.

3. Static routes, including primary and backup paths where needed, were added with ip route and checked using show ip route static.

4. RIP was activated using router rip, and directly connected networks were announced with network commands.

5. Connectivity was checked with ping and tracert, and communication between PC0 and PC1 was established successfully.

RESULT

Thus, static routing and RIP routing were configured and tested successfully using Cisco Packet Tracer.